

vision**2**learn

Coding Practices Level 3

Unit 2

Learning
outcomes



Unit 2 - Coding Design

Learning outcomes / Assessment criteria

1 Understand the use of coding standards

- 1.1 Identify the importance of coding standards
- 1.2 Explain the consequences of not following coding standards
- 1.3 Explain the processes and tools that could be used to promote coding standards

2 Understand the purpose of good coding principles and practices

- 2.1 Identify good coding principles and practices used by Software Developers
- 2.2 Explain why it is important to follow good coding principles and practices
- 2.3 Draw conclusions from the use of coding practices and principles

3 Know about coding reviews

- 3.1 Identify different types of coding review
- 3.2 Explain why code is reviewed
- 3.3 Explain why it is important to undertake a coding review
- 3.4 Explain the importance of effective feedback in a coding review
- 3.5 Distinguish the difference between submitting and performing a coding review
- 3.6 Explain the advantages and disadvantages of:
 - pair programming
 - mob programming

4 Understand databases

- 4.1 Explain what is meant by:
 - structured data
 - unstructured data
- 4.2 Describe what is meant by databases
- 4.3 Describe the structure and key components of a database table
- 4.4 Explore the use of primary and foreign keys
- 4.5 Describe techniques for querying databases
- 4.6 Explain the purpose of database indexing

5 Understand the principles of interface design

- 5.1 Identify the elements of UX (User Experience) design
- 5.2 Describe the differences between UX and UI (User Interface)
- 5.3 Explain the key skills required for creative problem-solving
- 5.4 Describe the key methods and techniques used in creative problem-solving
- 5.5 Explain the importance of information architecture and UX
- 5.6 Explain the principles of good user interface design

6 Understand the importance of building security by design into software at the development stage

- 6.1 Describe what is meant by building security by design into software at the development stage
- 6.2 Explain methods used to ensure software is secure
Identify types of security issues and threats that can impact software development